Type 1: Research Project

AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Real-Time Cyber-Physical Threat Detection for Industrial Control Systems

Team Number: [33]

S.No	Register Number	Student Name
1	CB.SC.U4CSE23003	Amarthya Sujai
2	CB.SC.U4CSE23020	Dwarakesh Venkatesh
3	CB.SC.U4CSE23049	Shyam Sundar Raju
4	CB.SC.U4CSE23050	Swetha C
5	CB.SC.U4CSE23054	Sudharsana Saravanan S

Guide: **Dr. Anantha Narayanan V**

Associate Professor, Department of Computer Science and Engineering

Industrial Control Systems (ICS)

A specialized category of CPS for critical infrastructures (Energy, Oil & Gas, etc).

- **Core Components:** PLCs, SCADA, RTUs, and HMIs.
- **Function:** Provides centralized visibility and distributed control of physical assets.

Historical Context

- Focus: Availability & Safety.
- Security: Minimal to none.
- Protocols: Proprietary & Isolated.

Modern Development (IT/OT Convergence)

- Focus: Connectivity & Remote Access.
- Risk: Expanded Attack Surface.
- Protocols: Standard IP-based (Vulnerable).

2. Uses of ICS in Critical Infrastructure

- Oil & Gas Industry
  - Drilling and well control (pressure, temperature, vibration)
  - Blowout Preventer (BOP) safety systems
  - Pipeline monitoring and leak detection via SCADA
  - Compressor and pump station automation
- Water Treatment Plants
  - Monitoring water levels, flow rates, and pressure
  - Automated chemical dosing (chlorine, pH control)
  - Pump and valve control for distribution systems
  - Detection of contamination or sensor anomalies
- Power Grid / Energy Sector
  - Substation automation and protection relays
  - Load balancing and frequency regulation
  - Turbine, generator, and transformer monitoring
  - Fault detection and grid stability control

3. Literature Survey: Comparative Analysis

Review of Key Approaches in ICS Security

Paper / Focus	Advantages	Limitations
Karacayılmaz et al. *AI-based IIoT* [1] (2024)	Adaptive to new threats; high accuracy using advanced AI models.	"Black-box" nature lacks explainability; high computational cost for legacy PLCs.
Murugesan et al. *Statistical Approach* [2] (2024)	Handles missing data effectively; mitigates early attack discrimination.	Limited to smart grids; not applicable to broader ICS environments.
Mohammed et al. *Field flooding* [3] (2023)	Specifically tailored for O&G infrastructure; effective against DoS.	Narrow scope (flooding only); may miss subtle Logic Injection attacks.
Barsha & Hubballi *State Transition*[4] (2024)	Detects semantic attacks by validating the *sequence* of operations.	Manual state modeling is complex; struggles with high-speed, concurrent states.
Athar et al. *Dual-Layer Attestation* [5] (2025)	Strong integrity verification; prevents firmware tampering.	High overhead for real-time loops; requires specific architecture changes.
Srivastava et al. *DDoS in SCADA* [6] (2024)	Critical protection against Availability attacks (DDoS) ensuring uptime.	Focuses on Network Layer (volumetric); misses Application Layer (logic) attacks.

Literature Survey: Analysis & Direction

- **Survey Scope:** We analyzed **6 key papers** focusing on AI-based detection, Statistical approaches, and Modbus protocol & SCADA vulnerabilities in Industrial Control Systems (ICS).

- **Research Gap:** Existing solutions typically suffer from one of three limitations:
  1. **Computational Latency:** Advanced AI or cryptographic methods introduce processing delays that violate the strict real-time constraints of safety-critical loops.
  2. **Narrow Focus:** Focuses only on specific attacks (e.g., Flooding) while ignoring subtle Logic/State manipulation.
  3. **Implementation Barrier:** Require expensive hardware upgrades that cannot be easily retrofitted to legacy systems.

4. Problem Statement

Industrial Control Systems (ICS) in critical sectors like Oil & Gas and Water Treatment operate under strict real-time and safety constraints yet face increasingly sophisticated cyber threats. Traditional security solutions often introduce latency, require significant resources, or fail to detect stealthy and zero-day attacks. This challenge is intensified by legacy PLCs and SCADA devices with limited CPU and memory. As breaches can cause serious safety, operational, and financial consequences, there is a need for a lightweight, real-time anomaly detection system capable of operating within minimal resource limits, identifying unknown threats, and incorporating physics-aware analysis.

5. SCADA Vulnerabilities & Threats

Critical Weaknesses

1. Man-in-the-Middle (MitM) [1]

SCADA systems, especially in substations, often communicate via unencrypted protocols.

- **Threat:** Attackers intercept and modify control signals in transit.
- **Impact:** Valid commands are replaced with malicious ones without the operator's knowledge.

2. Architectural Flaws [3]

Many legacy architectures lack internal segmentation (flat networks).

- **Threat:** Once an attacker breaches a single node (e.g., HMI), they have unrestricted access to all connected PLCs.
- **Impact:** Rapid propagation of malware across the entire facility.

3. State Manipulation [1, 5]

Attackers can decouple the HMI display from physical reality.

- **Threat:** Reporting a "Safe/Closed" state to the operator while the valve is actually "Open."
- **Impact:** Operator blindness leading to physical damage (e.g., Stuxnet centrifuge attack).

4. Digital Twin Desynchronization [5]

Modern ICS uses Digital Twins for predictive monitoring.

- **Threat:** Attackers feed false data to the Twin to ensure it matches the "expected" behavior.
- **Impact:** Hides the malicious activity from AI/ML detection systems relying on the Twin.

6. Historical Cyber Attacks: Key Incidents

Year	Attack Type	Target System	How They Attacked	Key Defenses
2021	Remote PLC manipulation	Florida water treatment plant (Oldsmar)	Attacker accessed plant via exposed remote access software (TeamViewer).	Disable unnecessary remote access, enforce MFA, network segmentation, continuous monitoring of PLC logic changes. [7]
2022	Industrial malware (ICS wiper)	Ukrainian energy infrastructure	Industroyer2 malware deployed against high-voltage electrical substations, issuing malicious commands.	Strict IT–OT segmentation, application allow-listing on engineering workstations, monitoring IEC-104 traffic. [8]
2023	Supply chain (backdoored SW)	Pipeline monitoring / SCADA	Compromised vendor updates inserted backdoor for remote SCADA access.	Signed/verified updates, strict OT change mgmt, zero-trust for servers/SCADA. [9]
2024	Zero-day exploit	Offshore platform control systems	Remote exploit of zero-day vuln for high-priv access & process control influence.	Harden/allow-list hosts, segment OT networks, OT anomaly detection. [10]
2025	Insider threat	Refinery control systems	Insider modified ICS settings, causing near-miss safety event.	Enforce least privilege and RBAC, apply UEBA monitoring, implement insider-risk programs. [11]

7. Industrial Visit to Roots Cast

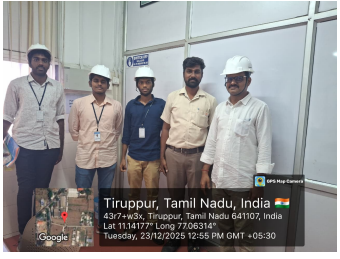
Key Insights Related to Security & Operational Integrity

Location: Roots Cast Private Limited, Arugampalayam, Coimbatore, Tamil Nadu 641017
Date of Visit: December 23, 2025

Key Observations:

- Real-Time Data Necessity
- Security through Network Isolation
- Role-Based Access Control
- On-Premise Data Processing
- Centralized Plant Visibility
- Latency in IIoT Communication





8. Proposed Methodology: Hierarchical State-Based Monitoring

Dual-Layer Global Physical Validation

We propose a two-tier architecture that validates system integrity by correlating internal controller logic with global physical process laws.

Layer 1: Local State Model (PLC)

Focus: Logic Integrity & Sequencing.

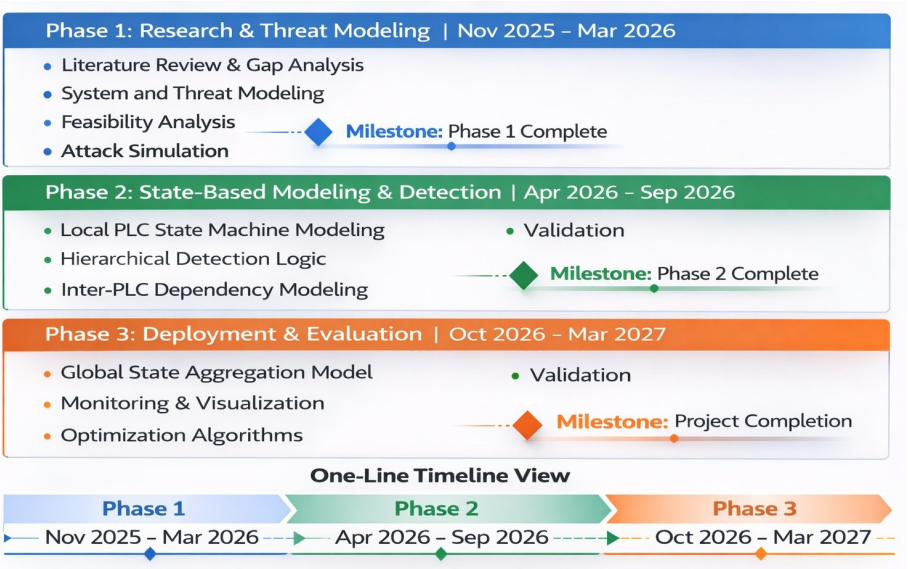
- Monitors the **internal transitions** of individual PLCs by analyzing the transitions of the PLC state.
- **Detection:** Flags "Illegal Transitions" (e.g., jumping from *Idle* to *High Pressure* without the *Ramping* state).
- Catches code-level tampering and unauthorized logic changes.

Layer 2: Global Model (Coordinator)

Focus: Process Physics & Cross-PLC Correlation.

- Monitors the **relationship** between multiple assets (e.g., Pump vs. Tank).
- Aggregates all the local state models to monitor the overall workflow.
- **Detection:** Flags "Semantic Gaps" where PLC reports "Normal" but the physics (flow/pressure) indicate a spike.

9. Project Timeline



10. Practical Implementation: Modbus Protocol Exploitation

Baseline Understanding via Pymodbus Attack Simulation

Register Mapping & Interaction

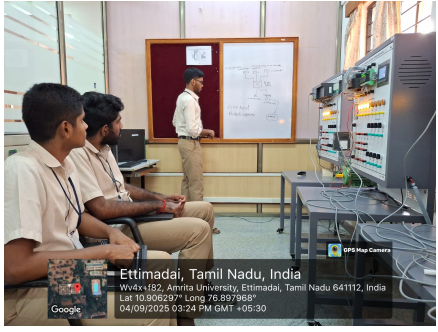
Using the `pymodbus` library, we developed a utility to interact with a target PLC:

- **Digital I/O:** Read/Write operations on *Coils* and *Discrete Inputs*.
- **Analog Data:** Manipulation of *Holding Registers* and *Input Registers*.

Attack Simulation

We simulated an **Unauthorized Packet Injection** attack:

- Forced LED and motors ON/OFF via register writing
- Altered motor speed to simulate equipment stress



Conclusion: Research Direction & Impact

Real-Time Cyber-Physical Threat Detection for ICS

The Challenge

Industrial Control Systems face evolving threats that current defenses cannot handle:

- **Real-time constraints**: Detection must not disrupt safety-critical operations.
- **Legacy hardware**: Minimal CPU/RAM in existing PLCs and SCADA systems.
- **Stealthy attacks**: Multi-stage compromises that evade traditional network-based IDS.
- **Critical impact**: Breaches in Oil & Gas, power, and water can cause catastrophic failures.

Expected Outcome

- Framework ready for adoption in Oil & Gas, water utilities, and power grids.
- Addresses critical research gap in real-time threat detection for legacy OT systems.
- Bridges advanced cybersecurity with practical ICS constraints.

References I

[1] G. Karacayılmaz and H. Artuner, "A novel approach detection for iiot attacks via artificial intelligence," *Cluster Computing*, vol. 27, no. 8, pp. 10 467–10 485, 2024. [Online]. Available: https://doi.org/10.1007/s10586-024-04529-w

[2] N. Murugesan, A. N. Velu, B. S. Palaniappan, B. Sukumar, and M. J. Hossain, "Mitigating missing rate and early cyberattack discrimination using optimal statistical approach with machine learning techniques in a smart grid," *Energies*, vol. 17, no. 8, 2024. [Online]. Available: https://www.mdpi.com/1996-1073/17/8/1965

[3] A. S. Mohammed, E. Anthi, O. Rana, N. Saxena, and P. Burnap, "Detection and mitigation of field flooding attacks on oil and gas critical infrastructure communication," *Computers Security*, vol. 124, p. 103007, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0167404822003996

[4] N. K. Barsha and N. Hubballi, "Anomaly detection in scada systems: A state transition modeling," *IEEE Transactions on Network and Service Management*, vol. 21, no. 3, pp. 3511–3521, 2024.

[5] S. O. Athar, M. N. Aman, and B. Sikdar, "Duatt: A dual-layer attestation scheme for plc-based industrial internet of things," *IEEE Internet of Things Journal*, vol. 12, no. 20, pp. 41 574–41 590, 2025.

[6] A. Srivastava, P. K. Saini, S. Tiwari, V. Sawan, and N. Garg, "Securing scada system from ddos attack," in *2024 2nd International Conference on Computer, Communication and Control (IC4)*, 2024, pp. 1–6.

[7] Federal Bureau of Investigation (FBI) and Cybersecurity and Infrastructure Security Agency (CISA), "Compromise of u.s. water treatment facility (oldsmar, florida)," mar 2021, joint Cybersecurity Advisory. [Online]. Available: https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-042a

References II

[8] ESET Research and CERT-UA, “Industroyer2: Industrial control system malware targeting ukrainian energy infrastructure,” apr 2022, analysis of Industroyer2 ICS attack on high-voltage substations. [Online]. Available: https://www.welivesecurity.com/2022/04/12/industroyer2-industroyer-reloaded/

[9] Waterfall Security Solutions, “Top oil and gas security challenges and best practices for protection,” 2023, oT Insights Center article. [Online]. Available: https://waterfall-security.com/ot-insights-center/ot-cybersecurity-insights-center/top-oil-and-gas-security-challenges-and-best-practices-for-protection/

[10] M. Khalil, “Zero-day exploit: A complete guide to threats & defense,” jun 2024, blog post; date adjusted to incident year for context. [Online]. Available: https://deepstrike.io/blog/zero-day-exploit-guide

[11] M. Schaefer, “The hidden danger: Insider threats in industrial control systems in 2025,” feb 2025, iCS Cybersecurity Weekly Summary. [Online]. Available: https://westoahu.hawaii.edu/cyber/ics-cybersecurity/ics-weekly-summaries/the-hidden-danger-insider-threats-in-industrial-control-systems-in-2025/

Thank You